Relational Algebra

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Relations

A relation R is defined as $R = (A, S)$ where A is a tuple of attributes and S is the set of rows in the relation.

✣ Formally:

$$
\begin{aligned}
R &= (A, S) \\
A &= (C_1, C_2, ..., C_{\text{cols}}) \\
C &= (\text{Name}, \text{Type}) \\
C_{\text{name}} &= C_1 \qquad \leftarrow \text{Useful Notation} \\
C_{\text{type}} &= C_2 \\
S &= \left\{ \left(s_{r_1}, s_{r_2}, ..., s_{r_{\text{cols}}} \right) \mid r \in 1..\text{rows} \right\}
\end{aligned}
$$

A C is an attribute, or Column definition, and it has a name and a type (read: set).

In R, attribute names are distinct:

$$\forall i, j : i \neq j \Rightarrow A_{i\ \text{name}} \neq A_{j\ \text{name}}$$

All values in S must match their respective attribute type:

$$\forall i \in 1...\text{cols}, s \in S : s_i \in A_{i\ \text{type}}$$

✣ Example:

Here is an example relation R_1:

Foo	Bar
1	2
3	4

$$
\begin{aligned}
R_1 &= (A, S) \\
A &= (('\text{Foo}', \mathbb{N}), ('\text{Bar}', \mathbb{N})) \\
S &= \{(1, 2), \\
&\quad\ (3, 4)\}
\end{aligned}
$$

Since S is a set, the order of $(1, 2)$ and $(3, 4)$ within make no difference.

Set Theoretic Operations

There are a few set theoretic operations that can be applied to relations R_1 and R_2:

- Union $\cup$
- Intersection $\cap$
- Difference or Subtraction $\setminus$ or $-$
- Cross Product $\times$

Let's define them all for $R_1 = (A_1, S_1), \quad R_2 = (A_2, S_2)$:

✣ $\cup, \cap, \setminus$

Let $\bigcirc = \{\cup, \cap, \setminus\}$ be a set of binaray set operations.

$R_1 \bigcirc R_2$ is defined iff $A_1 = A_2$.

$$R_1 \bigcirc R_2 = (A_1, S_1 \bigcirc S_2)$$

✣ Cross Product $\times$

The cross product works a little differently.

$R_1 \times R_2$ is only defined if there are no attribute names in common:

$$\begin{aligned} &\exists R_1 \times R_2 \text{ iff} \\ &\quad \forall x \in A_1 : \nexists y \in A_2 \ni : x_{\text{name}} = y_{\text{name}} \end{aligned}$$

If $R_1 \times R_2$ exists, then we do something like a cartesian product, but we merge the tuples together. Instead of $\{((a_1, a_2, ..., a_n), (b_1, b_2, ..., b_n))...\}$ we output $\{(a_1, a_2, ..., a_n, b_1, b_2, ..., b_n)...\}$.

$$\begin{aligned} R_1 \times R_2 = \Big(&\big({A_1}_1, {A_1}_2, ..., {A_1}_n, {A_2}_1, {A_2}_2, ..., {A_2}_m \big), \\ \text{New Set of Rows} \rightarrow \quad &\{(x_1, x_2, ..., x_n, y_1, y_2, ..., y_n) \mid (x, y) \in S_1 \times S_2\} \Big) \end{aligned}$$

Functional Dependencies